DATA JOURNALISM ON DEADLINE

WHAT IT'S REALLY LIKE TO PRACTICE "SOCIAL SCIENCE IN A HURRY"

My name is Ben

Sometimes I go by @palewire


All our slides are belong to
bit.ly/dataondeadline

Who am I?

Who am I?

Do you know?

Who am I?

Do you know?

You don’t. But you should.

I’m Philip Meyer.

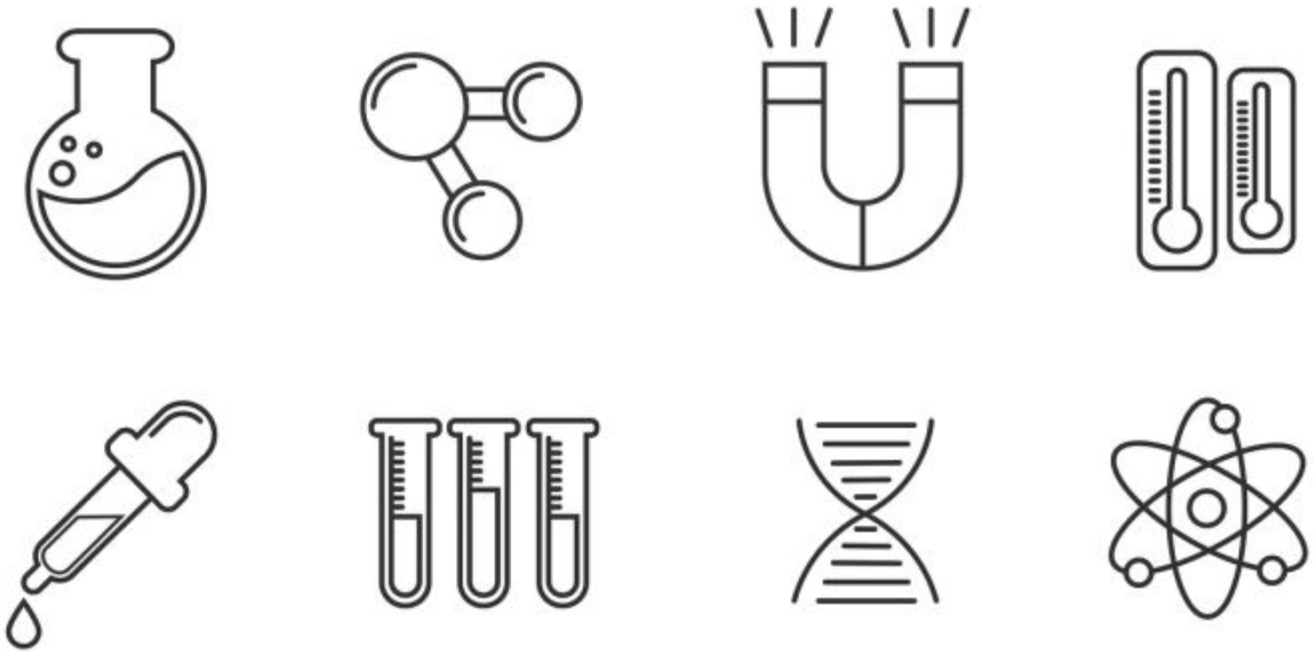
In 1973 I wrote the book “Precision Journalism,” crystalizing the ideas behind today’s “data journalism.”

You don’t have to take my word for it. Read the new history of data journalism, “Apostles of Certainty.” I’m all over it.

Why is Ben trotting out my esteemed image only to treat it like a debased meme? Because I once wrote the following:

To cope with the acceleration of social change in today’s world, journalism must become social science in a hurry.

IN THEORY



IN PRACTICE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1

U.S. is increasing nuclear power through uprating

By BY ALAN ZAREMBO AND BEN WELSH APR 17, 2011 | 12:00 AM

The U.S. nuclear industry is turning up the power on old reactors, spurring quiet debate over the safety of pushing aging equipment beyond its original specifications.
The little-publicized practice, known as uprating, has expanded the country's nuclear capacity without the financial risks, public anxiety and political obstacles that have halted the construction of new plants for the last 15 years.

The power boosts come from more potent fuel rods in the reactor core and, sometimes, more highly enriched uranium. As a result, the nuclear reactions generate more heat, which boils more water into steam to drive the turbines that make electricity.

Tiny uprates have long been common. But nuclear watchdogs and the U.S. Nuclear Regulatory Commission's own safety advisory panel have expressed concern over larger boosts — some by up to 20% — that the NRC began approving in 1998. Twenty of the nation's 104 reactors have undergone these "extended power uprates."


We’re flattering him here, but one generous way to describe the situation is to say our boy Ben has developed a *hypothesis*.

GIS Web Services for the FEMA National Flood Hazard Layer (NFHL)

FEMA's National Flood Hazard Layer

FEMA provides access to the National Flood Hazard Layer (NFHL) through web mapping services. The NFHL is a computer database that contains FEMA: s flood hazard map data. The simplest way for you to access the flood hazard data, including the NFHL, is through FEMAs Map Service Center (MSC). If you want to explore the current digital effective flood hazard data in a map, the best tool to use is the NFHL Viewer. From the NFHL Viewer, you may view, download, and print flood maps for your location

The data depict effective flood hazard information and supporting data used to develop the information. The primary flood hazard classification is indicated in the Flood Hazard Zones layer.

The NFHL layers include:

- Flood hazard zones and labels
- River Miles Markers
- Cross-sections and coastal transects and their labels
- Letter of Map Revision (LOMR) boundaries and case numbers
- Flood Insurance Rate Map (FIRM) boundaries, labels and effective dates
- Community boundaries and names
- Levees
- Hydraulic and flood control structures
- Profile and coastal transect baselines
- Limit of Moderate Wave Action(LiMWA)

Beginning February 2019, Coastal Barrier Resources System (CBRS) boundaries will no longer be depicted on static, legacy Flood Insurance Rate Maps (FIRMs) issued by the Federal Emergency Management Agency (FEMA), but will be available through the National Flood Hazard Layer (NFHL) Viewer via the CBRS live map service from the United States Fish and Wildlife Service (USFWS). This change to the FIRM will not impact CBRS property determinations nor any applicable prohibitions. FEMA and the USFWS are working together to update how customers access the most up-to-date information on CBRS boundaries.

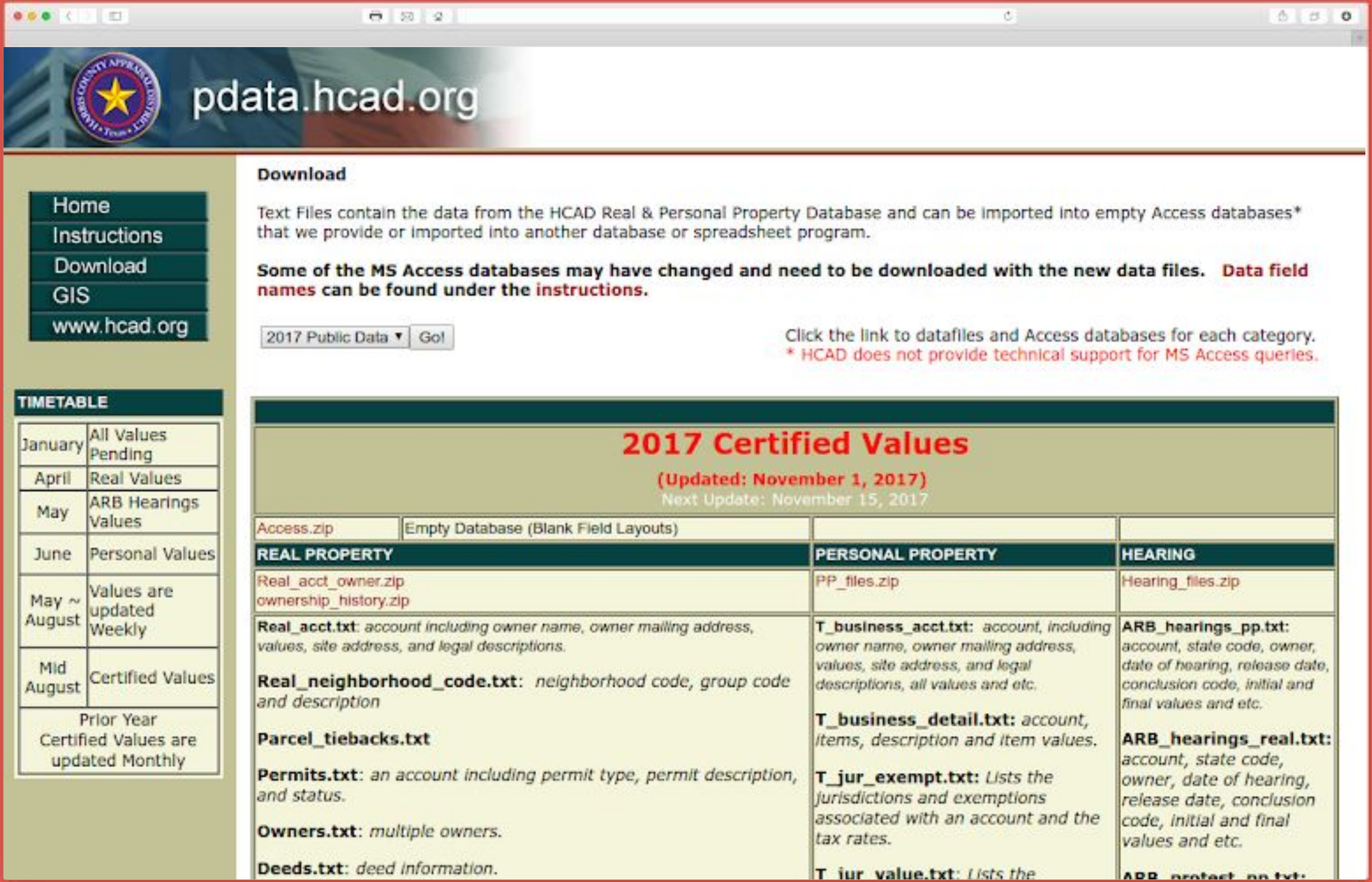
Not all effective Flood Insurance Rate Maps (FIRM) have geographic information system (GIS) data available. To view a list of available county and single-jurisdiction flood study data in GIS format and check the status of the NFHL GIS services please visit the NFHL Status Page.

Preliminary & Pending National Flood Hazard Layers

The Preliminary and Pending NFHL dataset represents the current pre-effective flood data for the country. These layers are updated as new preliminary and pending data becomes available, and data is removed from these layers as it becomes effective.

Preliminary & Pending Data

Preliminary data are for review and guidance purposes only. By viewing preliminary data and maps, the user acknowledges that the information provided is preliminary and subject to change. Preliminary data are not final and are presented in this national layer as the best information available at this time. Additionally, preliminary data cannot be used to rate flood insurance policies or enforce the Federal mandatory purchase requirement. FEMA will remove preliminary data once pending data are available. The preliminary flood hazard data is available through the preliminary FEMA Map Products page. If the preliminary flood hazard data search tool is unavailable, please use the alternate site. To explore the current digital preliminary flood hazard data in a map, the best tool to use is the Flood Map Changes Viewer (FMCV). From the Flood Map Changes Viewer, you may view, download, and print flood maps for your location.



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TIMETABLE

January	All Values Pending
April	Real Values
May	ARB Hearings Values
June	Personal Values
May ~ August	Values are updated Weekly
Mid August	Certified Values
Prior Year Certified Values are updated Monthly	

Download

Text Files contain the data from the HCAD Real & Personal Property Database and can be imported into empty Access databases* that we provide or imported into another database or spreadsheet program.

Some of the MS Access databases may have changed and need to be downloaded with the new data files. Data field names can be found under the instructions.

2017 Public Data ▾ Go!

Click the link to datafiles and Access databases for each category.
* HCAD does not provide technical support for MS Access queries.

2017 Certified Values

(Updated: November 1, 2017)

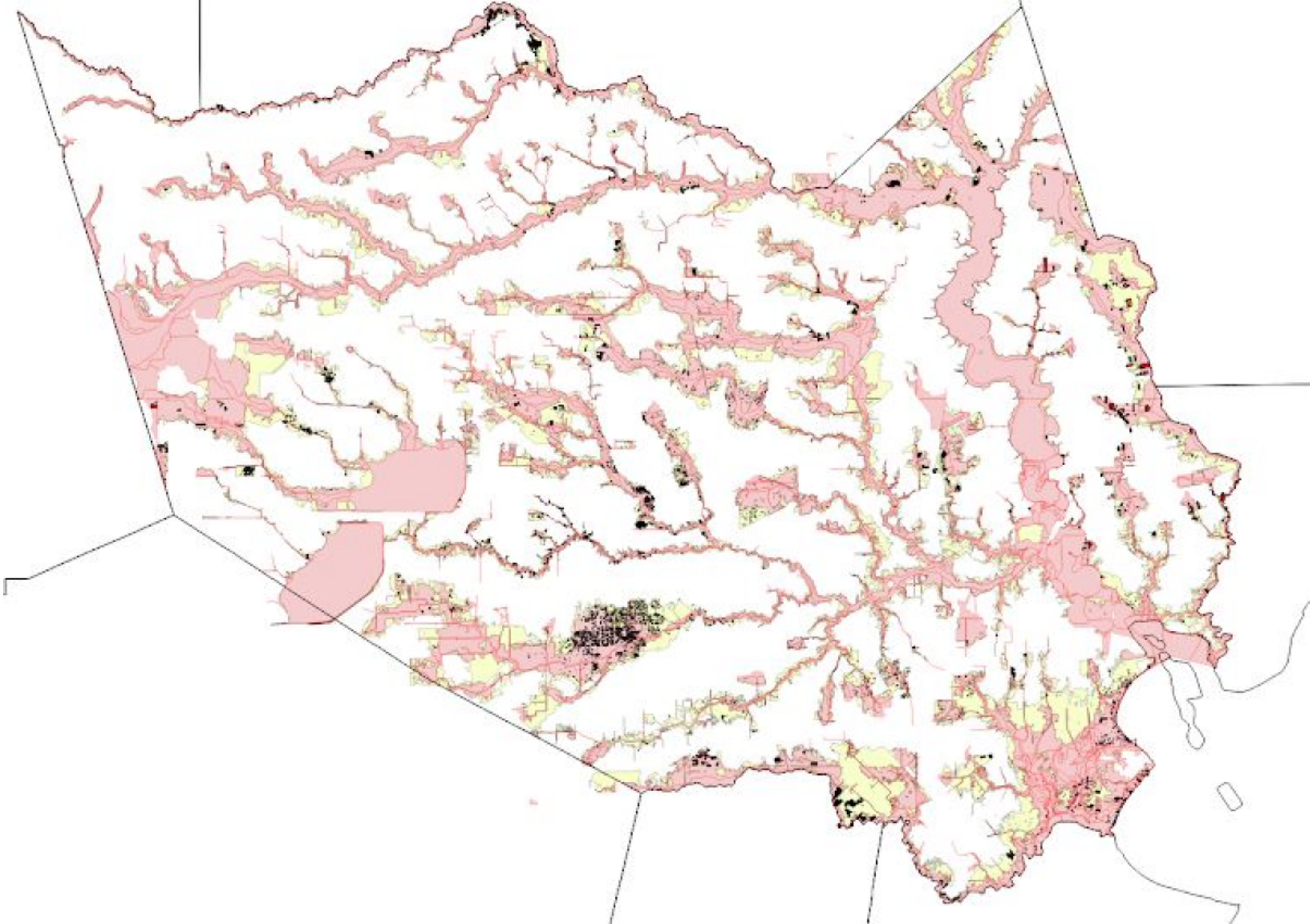
Next Update: November 15, 2017

Access.zip	Empty Database (Blank Field Layouts)		

REAL PROPERTY	PERSONAL PROPERTY	HEARING
Real_acct_owner.zip ownership_history.zip	PP_files.zip	Hearing_files.zip
Real_acct.txt: *account including owner name, owner mailing address, values, site address, and legal descriptions.* **Real_neighborhood_code.txt**: *neighborhood code, group code and description* **Parcel_tiebacks.txt** **Permits.txt**: *an account including permit type, permit description, and status.* **Owners.txt**: *multiple owners.* **Deeds.txt**: *deed information.*	**T_business_acct.txt**: *account, including owner name, owner mailing address, values, site address, and legal descriptions, all values and etc.* **T_business_detail.txt**: *account, items, description and item values.* **T_jur_exempt.txt**: *Lists the jurisdictions and exemptions associated with an account and the tax rates.* **T_jur_value.txt**: *Lists the*	**ARB_hearings_pp.txt**: *account, state code, owner, date of hearing, release date, conclusion code, initial and final values and etc.* **ARB_hearings_real.txt**: *account, state code, owner, date of hearing, release date, conclusion code, initial and final values and etc.* **ARB_protest_pp.txt**:

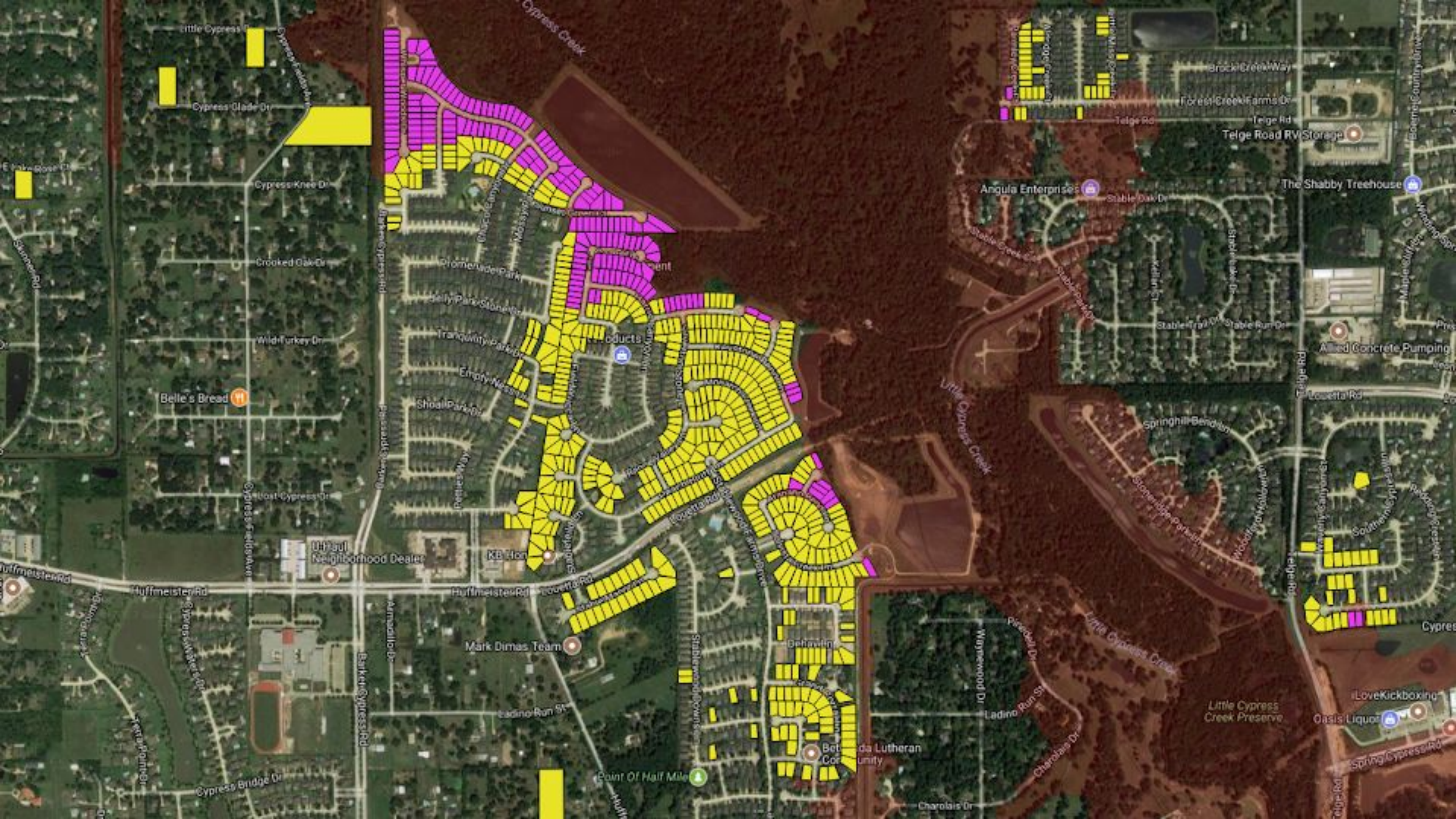
I admit it’s a stretch, but you might describe what our dunce has done as *operationializing* his research question


easy



I know what you’re thinking. Before anyone else is exposed to this flimsy work, we need some *peer review*

Okay, so if so-called theory is true, you should be able to *make predictions*, right?

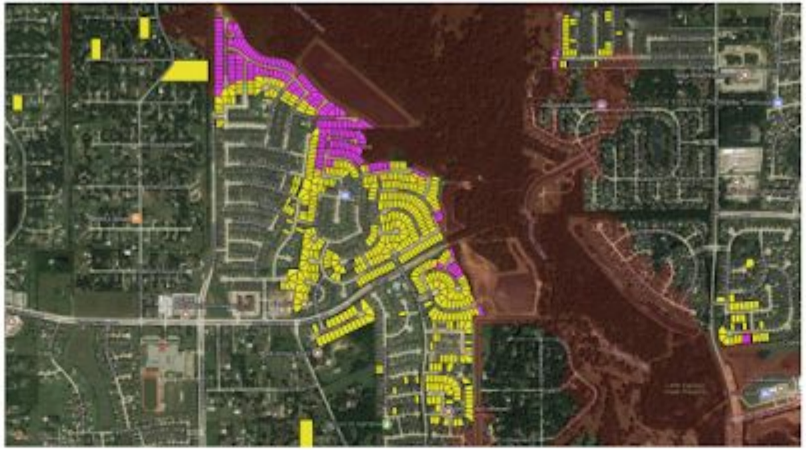


Little Cypress Creek
Little Cypress
Cypress Glade Dr
Cypress Knee Dr
Crooked Oak Dr
Wild Turkey Dr
Belle's Bread
Lost Cypress Dr
U-Haul Neighborhood Dealer
Huffmeister Rd
Promenade Park
Tranquility Park Dr
Empty Nest
Shoal Park Dr
Pebble Way
KB Hon
Louetta Rd
Mark Dimas Team
Ladino Run St
Cypress Bridge Dr
Point Of Half Mile
Bethesda Lutheran Community
Charolais Dr
Pinebell Dr
Waywood Dr
Little Cypress Creek Preserve
Brock Creek Way
Forest Creek Farms Dr
Telge Rd
Telge Road RV Storage
Angula Enterprises
Stable Oak Dr
The Shabby Treehouse
Stable Trail Dr
Stable Run Dr
Allied Concrete Pumping
Louetta Rd
Springhill Bend Ln
Stoneridge Parkway
LoveKickboxing
Oasis Liquor
Spring Cypress Rd
Barker Cypress Rd

New houses in Harris County flood zones

File Edit View Insert Format Tools Add-ons Help Last edit was on November 3, 2017

Little Cypress Creek



https://www.google.com/maps/@30.0004317,-95.6722853,2293m/data=!3m1!1e3

ID	Owner	Address	Notes
1319440030077	[redacted]	16834 RANGER RIDGE DR, CYPRESS	[redacted]
1319440030054		15826 WHISPER WOODS DR	
1301740050028		16511 SUNSET GREEN CT	

Spatially join buildings data with flood zones

Now we begin the analysis by importing our Python tools.

In [1]:
```
import os
import geopandas as gpd
import matplotlib.pyplot as plt
```

In [2]:
```
%matplotlib inline
```

Prepare utilities for opening files

In [3]:
```
get_output_path = lambda x: os.path.join("./output", x)
```

In [4]:
```
get_gdf = lambda x: gpd.read_file(get_output_path(x))
```

Because the source data files are so large, a single spatial join is difficult to execute. To work around the limitations of our available hardware, the following code splits up the shapefiles into chunks based on the year each building was built.

In [5]:
```
year_list = gpd.pd.np.arange(1800, 2017)
```

In [6]:
```
def split_year(df_name, func):
    """
    Split the provided dataframe into separate files using all the years in our dataset.
    """
    # Open the dataframe
    df = get_gdf(df_name)

    # Loop through all the years we're watching
    for year in year_list:

        # For each year pass the dataframe to the provided function (for filtering or analysis)
        year_df = func(df, year)

        # Write out the results
        if len(year_df) > 0:
            year_name = "{}-{}".format(year, df_name)
            year_path = get_output_path(year_name)
            year_df.to_file(year_path)

    # Pass back out the dataframe since it's already open. We might reuse it.
    return df
```

The master shapefile is read in and split into many pieces.

In [28]:
```
buildings = split_year("buildings-and-parcels.shp", lambda x, y: x[x.erected == y])
```

Each year's file is then spatial joined with the flood zones and output into a file with the matches.

In [7]:
```
def read_year(year, name_template="{}-buildings-and-parcels.shp"):
```

that construction should be prohibited in some flood-prone areas.

Since 2010, at least 7,000 residential buildings have been constructed in Harris County on properties that sit mostly on land the federal government has designated as a 100-year flood plain, according to a Washington Post review of areas at the greatest risk of flooding. Some other cities also allow building in flood plains, with varying degrees of regulation.

“Houston is the Wild West of development, so any mention of regulation creates a hostile reaction from people who see that as an infringement on property rights

FAIL.doc

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Paul F[redacted]
12119 ARBOR BLUE LN
[redacted]

Getting on a plane and doesn't want to talk.

There was parts of the neighborhood that was flooded. Different roads. On the main roads there was a lot of water. There was damages. There was reported tornado in the area. There are some people coming out to inspect the home. Not as bad as some other parts of the area. Of course I don't have their numbers. No not off the top. Not right now.

It is but it's a fairly new neighborhood but I think it was designed to properly handle it. No it wasn't necessarily the lake. The lakes was part of the runoff. He thinks they helped. There was some flooding on Bridgeland Lake Parkway.

Pope Elementary

Patty …

Ms. Cope, the principal.

Refused to answer any questions.

Super intendent of communications. Nicole R[redacted]

Tracy H[redacted]

Nobody in our neigborhood … REtention ponds were built by the Ag Corp of engineer. WE didn't even have any standing water on our street. But no houses in our neighborhood.

The A&M Corp of Engeineers built them. They drained the water down. You could see Cypress Lake and you could see it dropping water. It kind of all flowed. I think … no they didn't.

Keo L[redacted]

In order to qualify for federally backed mortgages, new homes in the 100-year flood zone must have insurance, which requires they be built above expected flood levels. Houston's 12-inch requirement, instituted four years after the first federal flood hazard maps came out, was an extra measure of protection.

Surrounding Harris County, which has an 18-inch requirement, has also undertaken a wide range of broader flood-control measures that help Houston. It requires builders to haul out at least as much dirt and concrete as they haul in to prevent filling in flood plains, and to create basins that hold runoff during storms.

Decade of data shows FEMA flood maps missed 3-in-4 claims

JADE BOYD – SEPTEMBER 11, 2017
POSTED IN: CURRENT NEWS

Rice, Texas A&M-Galveston study focused on 100-year maps for some Houston suburbs

An analysis of flood claims in several southeast Houston suburbs from 1999 to 2009 found that the Federal Emergency Management Agency's 100-year flood plain maps — the tool that U.S. officials use to determine both flood risk and insurance premiums — failed to capture 75 percent of flood damages from five serious floods, none of which reached the threshold of a 100-year event.

The research by hydrologists and land-use experts at Rice University and Texas A&M University at Galveston was published in the journal Natural Hazards Review just days before Hurricane/Tropical Storm Harvey inundated the Houston region and caused some of the most catastrophic flooding in U.S. history.


An aerial view shows extensive flooding from Harvey in a residential area in Southeast Texas, Aug. 31, 2017. (Air National Guard photo by Staff Sgt. Daniel J. Martinez)

"The takeaway from this study, which was borne out in Harvey, is that many losses occur in areas outside FEMA's 100-year flood plain," said study co-author Antonia Sebastian, a research associate at Rice's Severe Storm Prediction, Education and Evacuation from Disasters (SSPEED) Center and a postdoctoral researcher at Delft University of Technology in the Netherlands.

"What we've tried to show, both with this study and several others, is that it is possible to do better," said lead author Russell Blessing, a Texas A&M-Galveston graduate student with joint appointments at the SSPEED Center and Texas A&M-Galveston's Center for Texas Beaches and Shores. "There are innovative computational and hydrological tools available to build more predictive maps."

In the new study, Blessing, Sebastian and co-author Sam Brody, a professor of marine sciences at Texas A&M-Galveston, director of the Center for Texas Beaches and Shores and a SSPEED Center investigator, examined the Armand Bayou

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Houston's new poet laureate: 'Rice absolutely helped me get to this place'
17 May 2019 10:33 AM | No Comments

Walt Whitman once said, “I like the scientific spirit—the holding off, the being sure but not too sure...

… the willingness to surrender ideas when the evidence is against them: this is ultimately fine…

…it always keeps the way beyond open—always gives life, thought, affection, the whole man, a chance…

… to try over again after a mistake—after a wrong guess.”

Walt Whitman didn’t have to put out a newspaper.

MLB

Lance McCullers Threw 24 Straight Curveballs to Send the Astros to the World Series

By JEREMY WOO October 21, 2017

We knew Lance McCullers Jr. was nasty at his best, and he was at his *very* best on Saturday night, closing out the Yankees to send the Astros to the World Series.

What's crazy? He finished off the game, according to MLB.com's pitch tracking data...by throwing 24 consecutive curveballs to six different

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Power Rankings: Astros in the company of the '27 Yankees

MLB
Quarter-season

No other major metropolitan area in the U.S. has grown faster than Houston over the last decade, with a significant portion of new construction occurring in areas that the federal government considers prone to flooding.

But much of that new real estate in those zones did just fine, a Times analysis has found.

Though Houston has few zoning regulations, it requires that new homes be built at least 12 inches above expected flood levels. The 1985 regulation and others that followed proved widely effective in their biggest test to date — the record-setting rains of Harvey.

Of the nearly 39,000 single-family houses built within city limits since 2007, a total of 3,215 — or 8% — were erected in areas the federal government expects to flood every 100 years, and another 5,529 — or 14% — in areas expected to flood every 500 years, according to property records.

To look at damage to those homes, the Times chose five neighborhoods built inside a flood zone over the last decade and asked a sample of homeowners in each area how they fared. In case after case, owners of the newer homes rode out the hurricane safely.

Their stories are supported by the patterns that are emerging as the city assesses the damage.

The brunt of it appears to be borne by older houses — sometimes just doors away from newer, unscathed real estate — that predate federal and local anti-flooding regulations and account for 79% of the nearly 129,000 houses located in the federal flood zones.

Cool story bro.

But if you’re really doing science, you work needs to be *replicable*.

datadesk / houston-flood-zone-analysis Public archive

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palewire Update README.md · fa4ba21 on Oct 20, 2018 · 17 commits

.gitignore	First commit	5 years ago
01_download.ipynb	First commit	5 years ago
02_transform.ipynb	More markdown	5 years ago
03_merge.ipynb	More markdown	5 years ago
04_analysis.ipynb	Update 04_analysis.ipynb	5 years ago
README.md	Update README.md	4 years ago
requirements.txt	First commit	5 years ago

README.md

houston-flood-zone-analysis

The Los Angeles Times geospatial analysis of Houston property records conducted for the Nov. 8, 2017 story "How Houston's newest homes survived Hurricane Harvey".

The Times' findings are reproducible in the "analysis" notebook in this repository.

About

A Los Angeles Times geospatial analysis of Houston homes after Hurricane Harvey

www.latimes.com/projects/la-na-houst...

python · news · jupyter-notebook · pandas · data-journalism · data-analysis · geopandas · geospatial-analysis

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6 forks

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Jupyter Notebook 100.0%

Hurricane Harvey damaged more than 204,000 homes and apartment buildings in Harris County, almost three-quarters of them outside the federally regulated 100-year flood plain, leaving tens of thousands of homeowners uninsured and unprepared.

The new details come from the most extensive disclosure of flood data yet released by city and county officials. The numbers follow a pattern: More than 55 percent of the homes damaged during the Tax Day storm in 2016 sat outside the 500-year flood plain, as did more than one-third of those during the Memorial Day floods in 2015.

The findings raise questions about local government plans to prevent flooding that focus on tightening building codes inside the flood plain. They also cast further doubt on the accuracy of the maps used to identify housing most in danger of flooding.

No corrections is good.

What do you got for me today, though?